

Hallucination in Large Language Models: A Survey of Detection, Evaluation, and Mitigation

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Abstract

Large language models (LLMs) support a wide range of language tasks, but hallucinations remain a major obstacle to their reliable use. Their outputs may contradict established facts or introduce information unsupported by the available evidence. This survey presents an overview of hallucination in text-based LLMs, bringing together its definitions, detection methods, evaluation practices, and mitigation strategies. We begin by distinguishing world factuality from source faithfulness and illustrating common error types. We then examine detection approaches based on external evidence, sampling consistency, and internal model signals, alongside representative benchmarks and evaluation methods. We further discuss potential causes of hallucination and review interventions involving training, retrieval, decoding, and response revision. Through these comparisons, we examine the strengths and limitations of existing approaches and the trade-offs among factual accuracy, informative coverage, and computational cost. Finally, we discuss open challenges in evidence reliability, independent verification, and generalization, outlining directions for more reliable language generation. A companion repository of relevant literature and supporting materials is available at <https://github.com/lixincheng-xcl/LLM-Hallucination-Survey>.

1 Introduction

A fluent answer from a large language model (LLM) is not necessarily justified. In question answering (QA), a model may repeat a misconception; in summarization, it may add a plausible detail absent from the source. Both are called hallucinations, although their correctness criteria differ: source-unsupported content can be factually correct (Lin et al., 2022; Cao et al., 2022). Collapsing these settings obscures what a system improves.

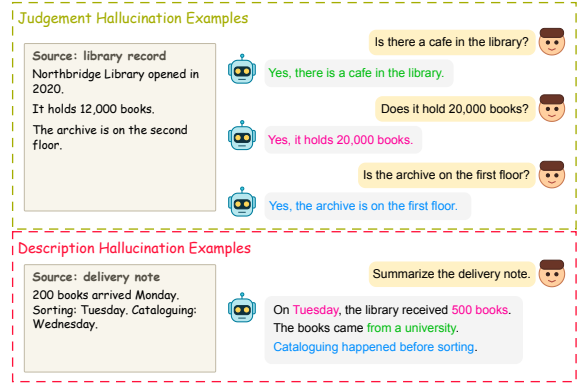


Figure 1: Illustrative hallucination examples in source-grounded question answering and summarization. Green, pink, and blue denote unsupported additions, attribute errors, and relation errors, respectively. Unsupported content is not necessarily false.

In Figure 1, the library record supports 12,000 books and a second-floor archive. The cafe assertion lacks support; the changed count and location contradict the record. The delivery summary also invents a donor and reverses events. Verification must specify the claim and its reference evidence.

Building on the broad factuality review of Wang et al. (2024), we connect each method’s evidence and access assumptions to the evaluation target it can support. This comparison distinguishes better correction from gains caused by withholding information or changing the verifier. Three questions organize the synthesis: **RQ1**, what reference frame defines an error; **RQ2**, what evidence and access enable detection; and **RQ3**, when does mitigation improve useful generation?

Scope and organization. This narrative survey covers 2022–2026 ACL-family research on text-based QA, long-form generation, summarization, dialogue, and retrieval-augmented generation (RAG), supplemented by mechanism-defining studies from other venues. Figure 2 maps concepts (§2), detection (§3), evaluation (§4), mitigation (§5), and



Figure 2: The structure of this survey. Green boxes list representative studies for each branch.

analysis (§6). Appendix A gives the selection criteria. Multimodal perception and general reasoning are outside the scope.

2 Concepts and Scope

2.1 Reference frames

World factuality concerns agreement with externally established facts; *source faithfulness* concerns support by supplied evidence. Neither entails the other: a response can repeat a false source faithfully or add a true but unlicensed detail (Cao et al., 2022; Adlakha et al., 2024). BEGIN evaluates attribution in grounded dialogue under this distinction (Dziri et al., 2022b).

Source-based labels should distinguish *supported*, *contradicted*, and *not established*, retaining the last separately from world falsity. World verification additionally needs a specified evidence source, date, and unresolved-claim policy. These are operational choices, not a universal definition. HalluLens revisits definitions and benchmark construction (Bang et al., 2025); comparisons must map study-specific labels before comparing scores.

2.2 Error relations and development

Entity, attribute, and relation errors are orthogonal to reference frames: a wrong date can contradict

either a document or a world fact. Annotation units also differ. ANAH provides sentence-level analysis, while HaluEval 2.0 extends empirical factuality analysis across domains (Ji et al., 2024; Li et al., 2024).

Research moved from distinct task-specific correctness targets in 2022 toward atomic verification and black-box self-checking in 2023 (Min et al., 2023; Manakul et al., 2023), followed by contextual monitoring and factuality alignment. Recent studies scrutinize judge reliability, information coverage, and temporal validity (Godbole and Jia, 2025; Samarinas et al., 2025; Jiang et al., 2026).

Three axes organize the comparison: reference frame, detection evidence, and intervention point. Benchmarks evaluate generated content or detectors; detectors can guide decoding. Fine-tuning and long-context studies motivate failures in supervision, knowledge access, and evidence use (Gekhman et al., 2024; Liu et al., 2024c). Across axes, we compare evidence access, generator signals, supervision, and cost, separating decision criteria from deployment requirements.

3 Detection: Evidence and Access

Detection estimates response, sentence, or span errors under a specified reference. Figure 2 groups

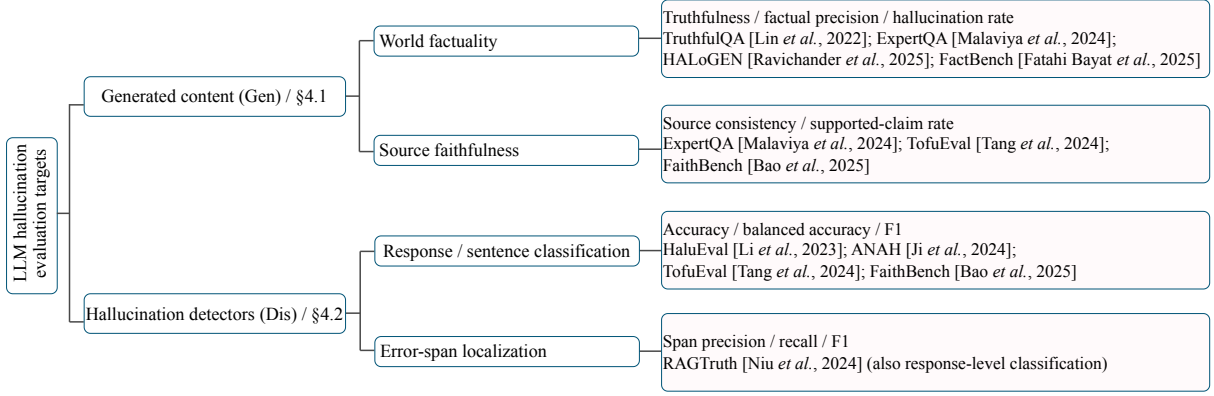


Figure 3: Taxonomy of hallucination evaluation methods for LLMs. The two branches distinguish generated-content evaluation from error-label prediction.

its evidence requirements: output access does not imply repeated samples, probabilities, or attention maps.

3.1 External evidence

Evidence-based methods compare outputs with supplied or retrieved knowledge. QAFactEval asks questions about summary content and checks answer recovery from the source; question generation and answerability affect its decisions (Fabbri et al., 2022). AlignScore learns alignment across heterogeneous tasks and aggregates source-chunk/output-sentence scores (Zha et al., 2023). Both support black-box generators but require trained auxiliary models. Question decomposition offers interpretable checks; learned alignment simplifies scoring. Neither eliminates verifier errors.

FActScore decomposes long answers into atomic facts and estimates their supported fraction relative to a designated source (Min et al., 2023). VeriScore instead extracts verifiable claims, using surrounding context to resolve references (Song et al., 2024). Extraction changes the score denominator; missing support differs from contradiction. PrefixNLI extends entailment to incomplete prefixes, enabling feedback during decoding at the cost of verification calls (Harary et al., 2026).

3.2 Sampling and consistency

SelfCheckGPT compares sentences with additional responses to the same prompt (Manakul et al., 2023). It avoids internal-access requirements but pays for repeated generation and consistency estimation. Agreement is only a proxy: samples can repeat a misconception, and comparison models can overlook disagreement.

Semantic uncertainty groups mutually entailing

answers before estimating entropy, reducing variation caused by paraphrases (Kuhn et al., 2023; Farquhar et al., 2024). The weighted estimator uses probabilities; a discrete approximation uses cluster frequencies. These methods particularly target unstable confabulations, whereas stable false beliefs need not produce high entropy. Sampling temperature, sample count, and equivalence judgments affect the estimate. Consistency can triage responses, but cannot replace independent evidence when justification is required.

3.3 Internal signals

FOCUS combines token probabilities with attention-based uncertainty propagation from an accessible proxy model (Zhang et al., 2023b). The target generator can remain black-box; the proxy must expose both signals. Hidden-state classifiers learn truth labels from activations (Azaria and Mitchell, 2023). Lookback Lens uses context-versus-generation attention features with a supervised linear classifier (Chuang et al., 2024a). These approaches can avoid repeated responses but require the specified model signals and, for probes, representative labels.

A separable activation pattern need not represent truth: associated falsehoods can resemble factual recall in controlled experiments (Cheang et al., 2026). Probes therefore need evaluation across natural error types and generators.

Alignment can expose a source contradiction but not establish source truth. Sampling tests claim reproducibility; probes test resemblance to labeled examples. Neither supplies independent factual evidence. Access requirements must distinguish the target generator, proxy, and verifier. Thus, “black-box” and “reference-free” are not interchangeable

deployment labels; evidence, signals, supervision, and inference budgets need separate specification.

4 Benchmarks and Evaluation

Figure 3 separates answer quality from detector performance. Table 1 instantiates this distinction; its sizes count different objects, not relative difficulty. Reference frame, annotation unit, and sampling process precede metric selection.

4.1 Generation evaluation

TruthfulQA evaluates misconception-oriented answers for truthfulness and informativeness, with a separate multiple-choice setting (Lin et al., 2022). ExpertQA distinguishes correctness from attribution through expert questions and claim-level judgments (Malaviya et al., 2024). HALoGEN uses domain-specific verifiers; FactBench combines user-derived prompts with web verification (Ravichander et al., 2025; Fatahi Bayat et al., 2025). Their scores are not interchangeable because evidence and verifier scopes differ.

TofuEval studies topic-focused dialogue summaries, where factuality must coexist with relevance and completeness (Tang et al., 2024). Precision needs coverage and refusal reporting, with explicit empty/unverifiable-answer policies.

Aggregation also changes the target. For $n > 0$ extracted claims and s supported claims, VeriScore combines $P = s/n$ with $R_K = \min(s/K, 1)$ as $F_1@K = 2PR_K/(P + R_K)$ for $s > 0$, and zero otherwise (Song et al., 2024). Its domain-specific $K > 0$ is the median extracted-claim count across evaluated responses. Unlike FActScore’s basic support fraction, this rewards supported-claim quantity as well as precision. However, R_K is not coverage of the information required by a question: extra supported details can raise it while leaving important aspects unanswered. Comparisons should fix K and separately audit extraction, verification, and task-relevant coverage. FactBench’s undecidable units additionally require an explicit aggregation policy (Fatahi Bayat et al., 2025).

4.2 Detector evaluation

HaluEval mixes generated task-specific examples with human-annotated general queries; recognition accuracy need not transfer to unselected errors (Li et al., 2023). ANAH annotates sentence-level types, evidence, and corrections; RAGTruth adds naturally generated RAG responses with error spans

(Ji et al., 2024; Niu et al., 2024). Response classification and span localization are different tasks: finding one error does not locate every unsupported claim.

The corpora in Table 1 supply annotations, but not necessarily every detector’s inputs. RAGTruth supports response and span testing; attention-based detection additionally requires compatible model traces, not merely source–response text (Niu et al., 2024). Feature extraction must specify the model and prefix construction; regenerating responses changes the evaluated outputs. Likewise, converting sentence scores into token labels adds a localization rule. TofuEval’s sentence and summary decisions require separate assessment (Tang et al., 2024).

FaithBench selects difficult cases through detector disagreement, making it a stress test rather than a prevalence estimate (Bao et al., 2025). TRIVIA+ examines long contexts and noisy supervision (Chen et al., 2026). Comparisons require matched splits, labels, generators, and evidence access, with thresholds fixed independently of test labels. Similar judge accuracy can conceal different error patterns and biased system-level estimates (Godbole and Jia, 2025). Mitigation must improve useful generation under an independently audited protocol.

5 Mitigation: Interventions and Assumptions

Figure 4 organizes interventions by where they act; selection also depends on what is trusted. Context-aware decoding (CAD) amplifies contextual evidence, whereas Astute RAG reconciles conflicting knowledge (Shi et al., 2024; Wang et al., 2025). Reporting a supplied record favors adherence; answering what is true may require rejecting that record. Neither policy dominates independently of the reference frame. Missing, ignored, and unreliable evidence therefore motivate different interventions.

5.1 Training and alignment

FaithDial supplies source-grounded dialogue revisions for generation and critic supervision (Dziri et al., 2022a). This requires curated examples. Self-Alignment instead derives preferences from a model’s self-evaluation, improves that evaluation through Self-Knowledge Tuning, and applies direct preference optimization (Zhang et al., 2024b). It reduces human labeling but depends on self-

Table 1: Hallucination evaluation benchmarks for LLMs. ‘Dis’, ‘Gen’, and ‘MC’ denote error-label discrimination, generated-content evaluation, and multiple choice, respectively; P and R denote precision and recall. Sizes count the stated units. Appendix B details the evaluation protocols.

Benchmark	Setting	Size	Focus	Metrics
TruthfulQA (Lin et al., 2022)	Gen/MC	817 questions	World factuality	Truthfulness; informativeness
HaluEval (Li et al., 2023)	Dis	35,000 samples	Mixed-task errors	Recognition accuracy
ANAH (Ji et al., 2024)	Dis	~12k sentences	QA support; contradiction	Type accuracy; F1
RAGTruth (Niu et al., 2024)	Dis	17,790 responses	RAG faithfulness	Response/span P, R, F1
ExpertQA (Malaviya et al., 2024)	Gen	2,177 questions	Correctness; attribution	Human factuality/support
TofuEval (Tang et al., 2024)	Dis/Gen	1,479 summaries*	Dialogue summaries	Balanced accuracy; error rate
HALoGEN (Ravichander et al., 2025)	Gen	10,923 prompts	Domain factuality	Atomic hallucination rate
FaithBench (Bao et al., 2025)	Dis/Gen	750 summaries	Summary faithfulness	Balanced accuracy; macro F1; conditional error rate
FactBench (Fatahi Bayat et al., 2025)	Gen	1,000 prompts	Verifiable claims	Hallucination score; precision

* Paper-reported retained count; the source arithmetic is inconsistent. HALoGEN’s count includes programming. FaithBench’s selected cases are not a prevalence sample. Dis/Gen indicates distinct possible uses, not a shared metric.

evaluation producing informative factuality preferences.

FactAlign uses sentence-level fine-grained Kahneman–Tversky optimization for credit assignment within mixed-quality answers (Huang and Chen, 2024). UAlign incorporates confidence and semantic entropy into a reward model before proximal policy optimization (PPO) alignment (Xue et al., 2025). RHIO targets contextual faithfulness: masking retrieval heads produces negative examples, control tokens distinguish faithful from unfaithful continuations, and contrastive decoding subsequently separates them (Huang et al., 2025b). FaithDial, Self-Alignment, and FactAlign thus differ in supervision source and granularity; RHIO additionally couples induced negatives with decoding. Their coverage depends on which facts supervisors can verify and which errors receive feedback. Table 7 compares access and resource requirements.

5.2 Retrieval and prompting

FLARE retrieves actively when provisional continuations signal uncertainty, while context-faithful prompting changes instructions without retraining (Jiang et al., 2023; Zhou et al., 2023). Retrieval adds information but introduces selection errors and evidence conflicts. Self-RAG trains retrieval and reflection tokens that control when to retrieve and assess relevance, support, and utility; candidate continuations can then be ranked using these signals (Asai et al., 2024). Astute RAG instead uses a training-free, source-aware consolidation proce-

dure: it elicits internal knowledge, compares it with retrieved passages, and iteratively resolves conflicts (Wang et al., 2025). This permits black-box use but adds calls and fallible reliability judgments. Comparison should separate retrieval quality, conflict resolution, and evidence use under matched corpora and budgets.

5.3 Decoding and internal intervention

CAD amplifies the difference between context-conditioned and context-free token distributions (Shi et al., 2024). For model parameters θ , next token y_t , context c , query x , prefix $y_{<t}$, and contrast strength $\alpha \geq 0$,

$$p_{\text{CAD}}(y_t) \propto \frac{p_{\theta}(y_t \mid c, x, y_{<t})^{1+\alpha}}{p_{\theta}(y_t \mid x, y_{<t})^{\alpha}}. \quad (1)$$

CAD requires logits and additional decoding work. It strengthens context adherence without assessing context truth. DoLa contrasts mature-layer and dynamically selected earlier-layer predictions within the same model (Chuang et al., 2024b). DoLa needs layer outputs rather than an external corpus, exposing parametric knowledge where CAD favors contextual evidence.

TruthX learns separate semantic and truth-related latent spaces and edits internal representations along a learned direction (Zhang et al., 2024a). It requires learning auxiliary components. Look-back Lens trains a classifier over context-versus-generation attention ratios and selects candidate continuations with that classifier (Chuang et al.,

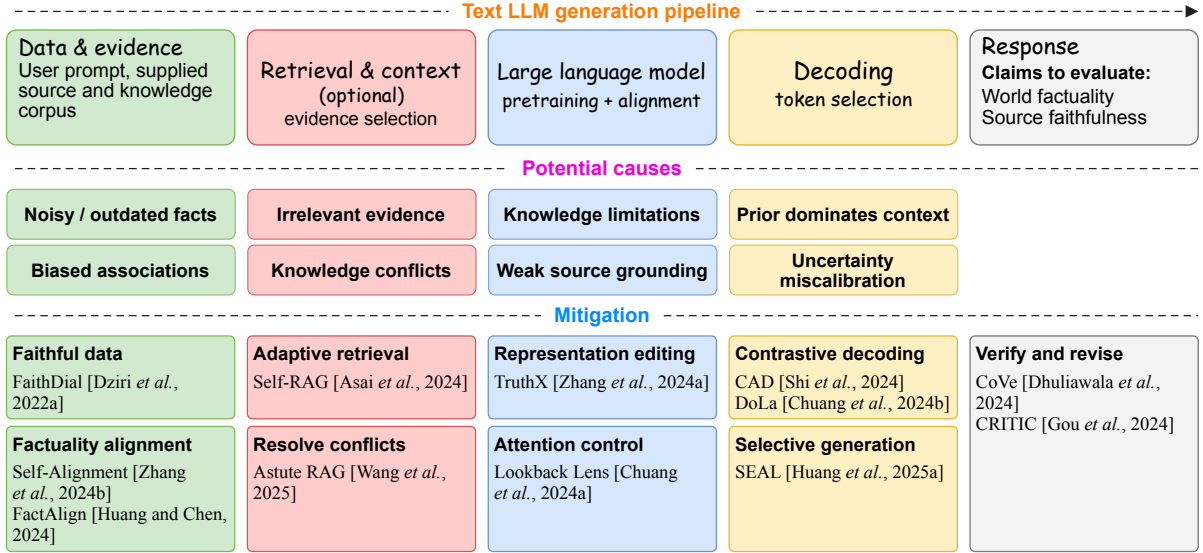


Figure 4: Potential causes and mitigation methods of hallucination in LLMs. Colors group stages of the generation pipeline; hybrid methods may span multiple stages.

2024a). Its mitigation closes the detector–generator loop, but also inherits detector errors. These mechanisms are unsuitable for text-only APIs without the required internal signals. Successful steering does not establish model-independent truth representations; transfer across architectures and natural errors needs testing.

5.4 Revision and abstention

Chain-of-Verification (CoVe) drafts an answer, generates verification questions, answers them independently, and revises the draft (Dhuliawala et al., 2024). Separating verification from the draft reduces answer contamination, but all stages can still share a parametric misconception. CRITIC adds tool feedback before revision, providing an external correction signal at the cost of tool calls and iterative generation (Gou et al., 2024). WikiChat also grounds dialogue through retrieval and verification, showing that these operations can be coordinated as a pipeline (Semnani et al., 2023). External feedback remains limited by search and tool reliability. Revision must preserve supported information while removing errors.

Abstention can reduce unsupported assertions. SEAL learns a rejection token during training and uses its probability to regularize decoding (Huang et al., 2025a); it is more specific than appending a refusal instruction. Lower error can reflect fewer answers. Refusal and revision therefore need coverage and information retention alongside precision.

6 Analysis and Future Directions

Reference frames determine whether an intervention succeeds. A response can follow an incorrect document faithfully or contradict a document while being factually correct. Consequently, gains from CAD and attention-guided decoding answer a different question from gains on world-knowledge QA. We propose evaluating both source support and world correctness on the same examples, retaining an unknown label when evidence is insufficient. A controlled retrieval experiment should vary document correctness and order while keeping questions and the generator fixed. Astute RAG motivates the conflict axis; Stable-RAG specifically addresses sensitivity to retrieval permutations (Wang et al., 2025; Zhang et al., 2026). Measuring both axes would distinguish stronger evidence use from more reliable evidence selection.

Confidence and internal signals are useful proxies, not truth certificates. Repeated agreement and confident internal activation can arise from a stable false association. Cheang et al. (2026) distinguish associated from unassociated hallucinations and find that the former resemble factual recall in the internal representations they study. This constrains broad interpretations of successful probing, uncertainty alignment, or representation editing: a detector may identify missing knowledge more readily than confidently recalled misinformation. Future evaluations should partition naturally generated errors by these mechanisms, hold out genera-

tor families, and test whether external verification improves detection specifically on confident falsehoods. Average accuracy conceals which failure process a detector recognizes.

Metric improvements must preserve useful information. Factual precision does not reward answering the difficult parts of a question. Controlled long-form experiments show that response length affects measured precision (Zhao et al., 2025), while ICAT explicitly measures the coverage of expected information aspects (Samarinas et al., 2025). Report unique supported claims, aspect coverage, and refusal alongside precision. Risk–coverage curves compare error rates at matched answered-question fractions. Revision additionally requires counts of corrected and newly introduced errors and retained supported claims; matched-length analysis tests whether gains mainly reflect less information.

Evaluation must remain independent of optimization. Godbole and Jia (2025) show that similar attribution-evaluator accuracy can conceal different false-positive/false-negative trade-offs and biased system-level error estimates. Optimizing and evaluating with the same verifier therefore risks rewarding its preferences. Independent auditing should report class-specific errors and test held-out domains and generators. Outdated reference facts also distort evaluation (Jiang et al., 2026), motivating versioned evidence and periodic label review. These requirements link Figure 4 back to Figure 3 and Table 1: improvements must survive independent checking under matched evidence, coverage, and budgets.

Resource trade-offs depend on the intervention stage. FactAlign pays for factuality feedback and parameter updates; Self-RAG combines offline supervision with online retrieval and candidate selection (Huang and Chen, 2024; Asai et al., 2024). CoVe and CRITIC instead add verification and revision calls during use (Dhuliawala et al., 2024; Gou et al., 2024). Training-free operation can still be costly; training expenditure may be amortized across queries but repeated after domain changes. Report both stages, including tokens, requests, latency, and memory.

Detection and mitigation require joint validation. An intervention can improve generation while making remaining errors harder to detect. Cheng et al. (2025) find that reasoning prompts

can reduce hallucinations while obscuring detection cues in their studied settings. For a fixed query distribution, let E denote an erroneous candidate and R the decision to release it. For $P(E) > 0$, the erroneous-release rate satisfies

$$P(E \cap R) = P(E)P(R | E). \quad (2)$$

No independence is assumed. Lower candidate error can be offset by higher false acceptance, so component improvements need not improve the complete system.

Proposed experiments should compare a baseline, each intervention, and their combination under matched questions, evidence, and budgets, crossing generators with fixed detectors. Report erroneous releases per query, error among released answers, and coverage, counting refusals as unreleased. Independently audit corrections, new errors, and retained supported claims at matched coverage to distinguish complementary correction from reduced answering. This tests Figure 4’s interventions against Figure 3’s targets; Appendix D details the protocol.

7 Conclusion

RQ1 requires separating world factuality from source faithfulness and declaring the claim unit and unknown-label policy. For **RQ2**, external evidence enables support checks, whereas sampling and internal signals estimate reliability under distinct access and supervision constraints. For **RQ3**, useful mitigation requires independently verified reliability gains, assessed jointly with informative coverage and computational cost. Future work should test detector–intervention combinations under evidence conflicts and distribution shifts, linking mechanism choice to residual errors.

Limitations

This narrative survey emphasizes English text tasks and ACL-family venues. Venue and keyword filters may omit relevant work; full-text checking was claim-specific rather than uniform. Heterogeneous generators, evidence, judges, and budgets prevent pooled effect estimates. The taxonomy and reporting protocol require prospective validation.

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A Selection, Provenance, and Reading Depth

This is a targeted narrative survey with a metadata snapshot of 15 September 2026, not an exhaustive systematic review or a meta-analysis. Discovery used official ACL Anthology collections for ACL, EMNLP, NAACL, EACL, Findings, and TACL across 2022–2026. Twenty-five collections were available; five requested year/venue combinations returned unavailable responses and were logged as such. A broad title filter for hallucination, factuality, faithfulness, attribution, uncertainty, retrieval, and related terms returned 1,766 records. Three targeted Anthology additions and seven external mechanism-defining studies produced 1,776 candidate records.

Selection prioritized direct relevance to text factuality or source faithfulness, distinct mechanisms, benchmark diversity, and evidence that qualified common assumptions. The working library contains 87 studies: 6 from 2022, 20 from 2023, 29 from 2024, 22 from 2025, and 10 from 2026. Eighty have Anthology records; the seven external records were checked against primary publication metadata. Canonical published versions were preferred over duplicate preprints. DOI and normalized-title checks found no exact duplicates among selected records; this is not a guarantee about near-duplicates in the reserve set. Automatic title triage excluded 401 candidates, and 1,288 remain in reserve without individual abstract assessment. These counts describe the actual workflow, not full-text exclusion decisions.

All 87 selected titles and abstracts were screened. Initial preparation inspected selected passages of 15 downloaded representative papers; drafting added targeted primary-source checks for the detailed detection, evaluation, and mitigation claims. This does not mean every selected paper was read in full. The companion evidence ledgers record claim-specific sections, access assumptions, and interpretive limits. The broad supplementary catalogue below distinguishes mechanisms without reproducing performance rankings. No experiments, independent benchmark replications, or statistical pooling were performed.

The synthesis cites 86 research studies from the 87-study discovery library, including references within the figures. Two additional surveys provide the visual-design sources: the layouts of Figures 1, 3, and 4 and Table 1 follow Liu et al. (2024a); the

structure tree in Figure 2 follows Liu et al. (2025). Figure 1 uses synthetic interactions to illustrate source-support distinctions. Figure 4 summarizes proposed mechanisms across studies rather than an experimentally identified causal decomposition.

Coding rules. For each study, record the correctness target, scoring unit, evidence source, model signals, supervision, and intervention stage. Assign detection branches by the evidence used for a decision; assign mitigation branches by the operation that changes generation. Keep separate entries for training and inference requirements. A method can occupy multiple branches: Lookback Lens uses attention for detection and candidate selection, while RHIO spans training and decoding. Record source-faithfulness targets separately from world-factuality targets, and distinguish target-generator access from proxy or verifier access. These rules make the comparison reproducible without forcing hybrid methods into exclusive categories.

B Benchmark Units and Protocol Caveats

Table 2 supplies the counting and interpretation details behind Table 1. A benchmark may support both a description of generated errors and evaluation of a detector, but those uses require different denominators and sampling assumptions. In particular, reporting error frequencies within deliberately selected difficult examples does not estimate a generator’s error prevalence on ordinary requests.

C Supplementary Comparative Evidence

Tables 3–6 extend the main synthesis with complementary studies. The final column states the inference boundary used in this survey; it is not an assertion that the original authors omitted every proposed test. These studies expand coverage of mechanisms and failure conditions rather than form a ranked list. Descriptive entries are grounded in the recorded primary metadata and selected source passages, with reading depth retained in the repository.

Table 2: Evaluation units, protocols, and limitations of the benchmarks in Table 1.

Benchmark	Unit, protocol, and limitation
TruthfulQA	The 817 questions target common misconceptions. Generative truthfulness/informativeness and multiple-choice answer selection are distinct settings. The original MC score measures normalized likelihood mass on true reference answers; it is not hallucination-detector classification.
HaluEval	The 35,000 examples combine task-specific generated examples with a human-annotated general-query portion. Artificial and naturally generated errors should be reported separately rather than treated as one deployment distribution.
ANAH	Approximately 12,000 annotated sentences provide error types, evidence, and corrections. Generative versus discriminative annotators describe detector implementations; their task remains prediction of annotation labels.
RAGTruth	The 17,790 responses come from 2,965 prompts and six generators, not 17,790 independent documents. Response classification and span localization have different matching rules and must be evaluated separately.
ExpertQA	The 2,177 expert-curated questions support factuality and attribution analysis. Correctness and cited-source support are separate judgments; exact evaluation subsets and answer-generation conditions matter.
TofuEval	Section 3.3 reports 1,500 initial summaries, removal of 23, and 1,479 retained summaries. Those values are arithmetically inconsistent. We retain the reported 1,479 rather than invent a corrected dataset size. Topic relevance and completeness are separate from factual consistency.
HALoGEN	The 10,923 prompts cover all nine domains, including programming. That is the complete-suite count, not a verified text-only subset size. Different domain verifiers do not constitute one universal human truth oracle.
FaithBench	The 750 summaries derive from 75 passages and ten models, selected using detector disagreement. This supports difficult-case detector analysis; error frequencies are conditional on selection, not population prevalence.
FactBench	The 1,000-prompt snapshot uses user-derived prompts and fine-grained verification. Supported, unsupported, and undecidable units require explicit aggregation rules. A dynamic benchmark needs a fixed version for reproducible comparisons.

Table 3: Additional benchmarks and evaluation protocols for hallucination in LLMs.

Study	Mechanism or evaluation design	Comparative implication and boundary
TRUE (Honovich et al., 2022)	Standardizes factual-consistency datasets across tasks and evaluates metrics at the example level.	Instance discrimination differs from system-level correlation. Combined existing tasks do not estimate real-world hallucination prevalence.
AggreFact analysis (Tang et al., 2023)	Stratifies summarization error annotations by generating model and error type; metric gains vary across strata.	Pooled scores can hide weaknesses on newer generators or rare error types.
BUMP (Ma et al., 2023)	Uses minimally differing human-written summary pairs with one introduced error.	A controlled error-sensitivity probe complements natural generation data; minimal pairs do not reproduce the full distribution of model errors.
ALCE (Gao et al., 2023)	Evaluates answer quality and citation quality in end-to-end retrieval and cited generation.	Citation presence, citation support, and factual correctness are distinct criteria. A plausible citation is not by itself evidence.
LongEval (Krishna et al., 2023)	Studies fine-grained human judgments and partial annotation for long summaries.	Fine-grained annotation requires agreement and coverage reporting; partial annotation does not exhaustively verify all claims.
FACTOR (Muhlgay et al., 2024)	Transforms factual corpora into true versus false continuation choices.	Controlled factual choice evaluates a different task from open-ended generation and error localization.
ReEval (Yu et al., 2024)	Perturbs evidence to create dynamic tests of evidence use under retrieval.	Static accuracy may reflect memorized knowledge rather than source adherence; counterfactual evidence requires an explicit faithfulness target.
Core (Jiang et al., 2025)	Filters redundant or uninformative subclaims before factual precision aggregation.	Trivial repetitions can inflate precision; informativeness and coverage are complementary criteria.
TRIVIA+ (Chen et al., 2026)	Adds long-context RAG detection data and controlled sample-dependent label noise.	Long-context and noisy-label tests complement evaluation on clean labels.
RGB (Chen et al., 2024)	Separates noise robustness, negative rejection, information integration, and counterfactual robustness in English/Chinese RAG.	Retrieval conditions define distinct RAG abilities, not one interchangeable hallucination score.

Table 4: Additional hallucination detection and explanation methods for LLMs.

Study	Mechanism or evaluation design	Comparative implication and boundary
WeCheck (Wu et al., 2023)	Aggregates weak labels over actual generated samples, then learns a noise-aware classifier.	Supervision quality and error distribution distinguish a learned consistency model from zero-shot entailment.
FactKB (Feng et al., 2023)	Uses knowledge-base factual pretraining with entity/relation objectives for factuality evaluation.	Structured supervision targets relation errors; broader transfer requires separate evidence.
TrueTeacher (Gekhman et al., 2023)	Uses LLM-labeled model-generated summaries to train a smaller consistency evaluator.	Distillation trades annotation cost for teacher-label dependence; generated errors differ from artificial perturbations.
FENICE (Scirè et al., 2024)	Extracts atomic claims and aligns them to a source using natural language inference (NLI).	Local alignment improves interpretability, while claim extraction and entailment both remain possible error sources.
FactReasoner (Marinescu et al., 2025)	Models evidence–claim logical relationships probabilistically after decomposition/retrieval.	Joint reasoning differs from independently verifying claims; its result still depends on evidence retrieval and relation estimates.
Future context (Lee et al., 2026)	Samples continuations and uses them as clues for black-box hallucination detection.	Generated continuations extend consistency checking without supplying independent external verification.
FaithLens (Si et al., 2026)	Trains detection and explanations using filtered synthetic data, supervised fine-tuning (SFT), and rule-based reinforcement learning.	Explanation quality and decision accuracy require separate validation; fluency alone does not establish explanation faithfulness.
QASemConsistency (Cattan et al., 2026)	Decomposes claims into predicate–argument QA pairs for fine-grained support localization.	Semantic error localization supplies a finer target than response-level binary classification.

Table 5: Studies of hallucination mechanisms and reliability in LLMs.

Study	Mechanism or evaluation design	Comparative implication and boundary
Factual confidence (Mahaut et al., 2024)	Compares confidence estimators and checks stability under meaning-preserving input changes.	Confidence can be unstable across paraphrases; access and supervision requirements differ.
New-knowledge fine-tuning (Gekhman et al., 2024)	Controlled closed-book QA experiments vary the amount of unfamiliar factual information in SFT.	The findings motivate knowledge-aware supervision within the studied fine-tuning setup.
Known facts (Jiang et al., 2024)	Examines layer-wise prediction dynamics when questions about the same fact yield correct versus incorrect answers.	Errors can reflect knowledge access rather than absence; measured patterns do not prove universal causal mechanisms.
Lost in the Middle (Liu et al., 2024c)	Evaluates how position of relevant evidence affects long-context use.	Evidence being present is insufficient for reliable use; its tasks are not themselves a comprehensive hallucination taxonomy.
Hidden-state limits (Servadio et al., 2025)	Tests factuality encoding on more realistic, model-dependent generated data.	Accuracy on synthetic true/false facts does not establish probe generalization to target-model outputs.
Unfamiliar supervision (Kang et al., 2025)	Relates hallucination response patterns to supervision of unfamiliar fine-tuning examples, including reward-model errors.	Factuality alignment depends on how unknowns and reward errors are labeled.
Illusion of Progress (Janiak et al., 2025)	Examines lexical-overlap labels and response-length heuristics in detector evaluation.	Semantic validation and simple baselines remain necessary; an LLM judge is not automatically a gold standard.
CoT obscures cues (Cheng et al., 2025)	Measures how reasoning prompts shift hidden/probability signals and detector effectiveness.	A generation intervention shifts the detector’s input distribution, making calibration transfer uncertain.
Hallucination tax (Song et al., 2025)	Studies refusal degradation after reinforcement fine-tuning, primarily on unanswerable mathematics, with reported transfer to factual QA.	Evidence is not a universal result for every factual-generation task.
Source preferences (Schuster et al., 2026)	Controlled synthetic-source study of source preferences and repetition in conflicting English evidence.	Frequency and source cues influence conflict resolution; synthetic-source preference is not proof of real-world source credibility.

Table 6: Additional hallucination mitigation methods for LLMs.

Study	Mechanism or evaluation design	Comparative implication and boundary
FaithfulRAG (Zhang et al., 2025)	Models fact-level conflicts between retrieved and parametric knowledge before response generation.	Explicit reconciliation complements source preference; source adherence and world correctness remain distinct.
CoCoA (Khandelwal et al., 2025)	Adapts token-level decoding using confidence/context distribution differences.	Adaptive and fixed contrast differ in conflict sensitivity; low-conflict behavior also matters.
Context-DPO (Bi et al., 2025)	Constructs preferences from faithful versus stubborn responses under knowledge conflict.	A context-faithfulness objective differs from truthfulness with respect to external facts.

D A Common Reporting Protocol

The following protocol is a synthesis of the comparisons in Sections 3–6, not a benchmark run. It can be used to evaluate whether an intervention in Figure 4 solves the kind of error illustrated in Figure 1 under the target in Figure 3.

D.1 Fix the target before selecting a metric

Specify whether the target is world correctness, source support, or both; the answer unit; and the treatment of unknown or unverifiable claims. Record the generator/version, prompts, retrieval corpus/date, evidence truncation, decoding parameters, and judge/version. Use the same evidence and question distribution across systems where possible. If a method changes retrieval, separate retrieval quality from generation conditional on the retrieved passages.

For a nonempty extracted claim set C and fixed evidence E , a schematic support precision is

$$P_{\text{support}}(C, E) = \frac{\sum_{c \in C} \mathbf{1}[E \models c]}{|C|}. \quad (3)$$

This expression describes an aggregation principle, not the complete implementation of every factuality metric. Extraction determines the denominator; duplicate, trivial, subjective, and undecidable claims need explicit handling. An empty answer is not assigned perfect useful factuality by this definition. FActScore and VeriScore motivate claim-level analysis, while Core highlights the role of informative subclaims (Min et al., 2023; Song et al., 2024; Jiang et al., 2025).

D.2 Separate detection, localization, and generation quality

For binary detection, report class prevalence, confusion counts, precision/recall/F1, and balanced accuracy, $\frac{1}{2}(\text{TPR} + \text{TNR})$, where appropriate. Fix thresholds on independent validation data. For

spans, specify tokenization and exact versus overlap matching. For continuous scores, report discrimination and calibration separately; neither establishes the truth of an individual answer. Inspect false-positive and false-negative examples across error types, evidence lengths, and generator families.

For generation, pair factuality with unique supported content and required-aspect coverage. A source-faithful answer may still omit the answer to the question. Human auditing should sample both high- and low-scoring outputs, use explicit instructions, and report disagreement or adjudication. LongEval provides relevant human-evaluation guidance (Krishna et al., 2023). Avoid pooling scalar scores whose claim extraction, evidence, or unknown-label policy differs.

D.3 Measure selection and revision effects

Let A_τ be the answers retained by threshold τ among N questions, and let e_i indicate an erroneous answer under the declared reference frame. For nonempty A_τ , define

$$\begin{aligned} \text{Coverage}(\tau) &= \frac{|A_\tau|}{N}, \\ \text{Risk}(\tau) &= \frac{\sum_{i \in A_\tau} e_i}{|A_\tau|}. \end{aligned} \quad (4)$$

Comparing risk at matched coverage prevents a system from appearing universally better merely by refusing more questions. This response-level selective-evaluation definition is distinct from SEAL’s token-level training objective. For revision, count corrected errors, newly introduced errors, retained supported claims, and required information lost. Measure both before and after revision on the same initial drafts.

D.4 Audit independence, robustness, and cost

A verifier used for training or candidate selection should not be the sole final judge. Use indepen-

Table 7: Access requirements and computational resources of representative methods. Resource descriptions are qualitative and do not imply measured cost rankings.

Family / example	Required access or supervision	Principal extra resource	Important boundary
Source alignment / AlignScore	Source and output text; trained auxiliary model	Verifier inference over chunks	Auxiliary-model errors and evidence truncation
Claim verification / FActScore	Claim extraction, knowledge source, retrieval and verifier	Search and per-claim checks	Retrieval failure differs from disproof
Self-checking / Self-CheckGPT	Repeated outputs; no generator internals	Additional generation and comparisons	Repeated falsehood can appear consistent
Semantic entropy	Samples and semantic equivalence; likelihoods for weighted estimation	Sampling and clustering	Estimator access differs by variant
Probe / Lookback Lens	Attention maps and labeled training spans	Feature extraction and probe inference	Supervised contextual proxy, not proof of truth
Factuality alignment / FactAlign	Generator training and sentence-level factual feedback	Verification plus optimization	Reward errors can be reinforced
Retrieval control / Self-RAG	Training, retrieval, reflection-token supervision	Training and retrieved candidates	Reflection judgments need auditing
Contrastive decoding / CAD	Context-conditioned and context-free logits	Extra distribution at decoding	Greater source adherence can amplify false evidence
Layer contrast / DoLa	Intermediate layer predictions	Layer-output computation	Parametric knowledge is not external evidence
Representation editing / TruthX	Hidden states and learned auxiliary components	Auxiliary learning and intervention	Transfer beyond training conditions is uncertain
Verification / CoVe, CRITIC	Model calls; CRITIC also needs tools	Verification and revision rounds	Isolated prompts are not independent knowledge
Selective generation / SEAL	Rejection-token training and regularized decoding	Training; altered answering behavior	Lower risk must be interpreted at matched coverage

dently collected labels or an independently audited evaluator. Test natural errors from held-out generators, difficult consistent falsehoods, evidence conflicts, context permutations, paraphrases, and updated facts. Disaggregate performance rather than interpreting an average as proof of transfer. Verify with Caution and temporal benchmark analysis motivate independent error inspection and timestamped evidence (Godbole and Jia, 2025; Jiang et al., 2026).

Table 7 summarizes method access and deployment trade-offs. Report offline annotation and training costs separately from inference calls, sampled tokens, retrieval requests, tool invocations, latency, and memory. A black-box method can be expensive, while an internal method can be unavailable even if computationally economical. Do not infer a fixed latency multiplier from the number of model distributions or stages without measurement. Comparisons should either match budgets or present a quality–cost trade-off.